

Machine Learning-based Late Delivery Risk Prediction in Global Supply Chain Operations

Research Paper

Abstract

Late deliveries in global supply chain operations result in customer dissatisfaction, service-level agreement (SLA) violations, operational inefficiencies, and increased logistics costs. Traditional logistics systems primarily respond to delivery delays after they occur, limiting the ability of organizations to proactively manage shipment risks. This research presents a machine learning-based predictive analytics framework for identifying late delivery risks before shipment execution in global logistics operations. The proposed system uses operational, geographical, financial, customer, and shipping-related features from a large-scale logistics dataset associated with APL Logistics under the KWE Group.

Comprehensive data preprocessing and feature engineering techniques were applied to improve predictive performance. Features such as Shipping Pressure Index, Order Complexity Score, Regional Risk Score, and Shipping Mode Risk were developed to capture operational patterns influencing delivery delays. Multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost were trained and evaluated using classification metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Experimental results demonstrated that XGBoost achieved the best predictive performance, effectively identifying high-risk shipments with strong recall and ROC-AUC scores. In addition, an interactive Streamlit dashboard was developed to provide real-time operational risk monitoring, explainable AI insights, and shipment prioritization capabilities. The proposed framework transforms traditional reactive logistics management into proactive, intelligent, data-driven supply chain operations.

1. Introduction

Global supply chain operations have become increasingly complex due to rapid growth in e-commerce, international trade, customer expectations, and logistics network dependencies. Organizations operating in transportation and logistics sectors must maintain high delivery performance while minimizing operational costs and ensuring customer satisfaction.

One of the major challenges faced by logistics organizations is late delivery risk. Shipment delays can occur due to multiple operational factors such as regional congestion, insufficient shipping capacity, poor route planning, unexpected disruptions, high order complexity, and inefficient resource allocation.

Late deliveries create several business problems including:

- Service-level agreement (SLA) breaches
- Increased operational costs

- Customer dissatisfaction
- Brand reputation damage
- Loss of repeat customers
- Emergency rerouting expenses
- Reduced operational efficiency

Traditional logistics systems mainly focus on analyzing historical delays after they occur. However, modern supply chain operations require predictive intelligence capable of identifying risky shipments before execution. Early identification enables organizations to proactively manage operations through shipment prioritization, route optimization, resource allocation, and customer communication.

This research focuses on developing a machine learning–based predictive framework capable of estimating the probability of late deliveries before shipment execution. The project uses logistics operational data associated with APL Logistics to build predictive models, operational dashboards, and explainable AI systems for supply chain risk intelligence.

2. Problem Statement

APL Logistics faces operational challenges in identifying high-risk shipments before delays occur. Existing operational systems mainly react to delays after shipment execution, resulting in costly last-minute interventions and inefficient logistics management.

The organization lacks:

- A predictive system to identify late-delivery risks
- Quantitative risk scores for shipment prioritization
- Explainable operational insights into delay drivers
- Intelligent dashboards for proactive logistics monitoring

Therefore, a machine learning–based predictive framework is required to estimate delivery risk probability and support proactive operational decision-making.

3. Objectives

3.1 Primary Objective

Develop a machine learning system capable of predicting whether a shipment is likely to experience late delivery.

3.2 Secondary Objectives

- Generate delivery risk probability scores
- Classify shipments into risk categories
- Identify key operational drivers of delay
- Build explainable AI insights for operational trust
- Develop a Streamlit dashboard for logistics monitoring

- Support proactive supply chain decision-making
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4. Literature Review

Machine learning and predictive analytics have become increasingly important in supply chain management and logistics optimization. Several studies have demonstrated the effectiveness of predictive models for transportation risk analysis, shipment forecasting, and logistics optimization.

Previous research has applied techniques such as Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, and Neural Networks to supply chain forecasting problems. Researchers have also emphasized the importance of explainable AI in operational systems to improve decision transparency and managerial trust.

Existing studies highlight several major factors contributing to shipment delays:

- Shipping mode inefficiencies
- Regional congestion
- Inventory limitations
- High order complexity
- Transportation disruptions
- Inefficient logistics scheduling

However, many traditional systems focus primarily on historical analytics rather than proactive predictive intelligence. This research contributes by integrating predictive machine learning, operational risk scoring, explainable AI, and real-time dashboard analytics into a unified logistics intelligence system.

5. Dataset Description

The dataset contains operational logistics information related to shipping, customer details, financial attributes, regional data, and product characteristics.

5.1 Important Dataset Features

Feature Name	Description
Shipping Mode	Shipping method used for delivery
Days for shipment (scheduled)	Planned shipment duration
Sales	Total sales amount
Order Item Quantity	Number of units ordered
Customer Segment	Customer category
Market	Global market region

Feature Name	Description
Order Region	Regional delivery area
Product Price	Selling price of product
Order Profit Per Order	Profit earned from order
Late_delivery_risk	Target variable

5.2 Target Variable

The target variable used in this research is:

Late_delivery_risk

Where:

- 1 = Late Delivery
- 0 = On-Time Delivery

6. Research Methodology

The proposed framework follows a structured machine learning pipeline consisting of:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Exploratory Data Analysis
5. Model Development
6. Model Evaluation
7. Explainable AI Analysis
8. Dashboard Development

6.1 Workflow

Data Collection ↓ Data Cleaning ↓ Feature Engineering ↓ Train-Test Split ↓ Machine Learning Models
↓ Performance Evaluation ↓ Explainable AI ↓ Streamlit Dashboard

7. Data Preprocessing

Data preprocessing is essential to improve data quality and ensure accurate model training.

7.1 Missing Value Handling

Missing numerical values were replaced using median imputation, while categorical missing values were replaced using mode imputation.

7.2 Duplicate Removal

Duplicate records were removed to avoid data redundancy and biased learning.

7.3 Encoding Techniques

Categorical features were transformed using:

- One-Hot Encoding
- Label Encoding

High-cardinality features such as Product Name and Customer City were encoded using Label Encoding.

7.4 Feature Scaling

Numerical features were standardized using StandardScaler where necessary.

7.5 Class Imbalance Handling

The dataset exhibited imbalance between delayed and non-delayed shipments. SMOTE and class-weight balancing techniques were used to address this issue.

8. Feature Engineering

Feature engineering was performed to improve predictive capability by extracting operational intelligence from raw logistics data.

8.1 Shipping Pressure Index

This feature measures operational pressure based on order quantity and scheduled shipment duration.

Shipping Pressure Index = Order Quantity / Scheduled Shipment Days

Higher values indicate increased operational burden and potential delay risk.

8.2 Order Complexity Score

This feature measures complexity based on quantity and product pricing.

Order Complexity Score = Quantity × Product Price

Complex orders typically require additional coordination and logistics effort.

8.3 Profit Margin

Profit Margin = Order Profit / Sales

This feature captures business profitability patterns associated with shipment risk.

8.4 Regional Risk Score

Historical delay rates by region were used to generate congestion-based regional risk indicators.

8.5 Shipping Mode Risk

Historical late-delivery frequencies for each shipping mode were used to estimate mode-based risk.

8.6 High-Value Order Flag

Orders exceeding the 75th percentile sales threshold were classified as high-value orders.

8.7 Bulk Order Flag

Orders exceeding the 75th percentile quantity threshold were classified as bulk orders.

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand shipment patterns, delay distributions, and operational risk trends.

9.1 Delay Distribution Analysis

The dataset showed a significant proportion of delayed deliveries, indicating operational inefficiencies in shipment management.

9.2 Shipping Mode Analysis

Standard shipping methods exhibited higher delay frequencies compared to express and same-day shipping modes.

9.3 Regional Risk Analysis

Certain regions demonstrated consistently higher congestion and delivery delay rates.

9.4 Customer Segment Analysis

Corporate and bulk-order customers exhibited relatively higher operational risk due to order complexity.

9.5 Financial Analysis

Large discounts and low profit margins showed correlations with operational inefficiencies and shipment delays.

10. Machine Learning Models

Multiple machine learning algorithms were implemented and compared.

10.1 Logistic Regression

Logistic Regression was used as a baseline interpretable model for binary classification.

Advantages:

- Simple and interpretable
- Fast training
- Suitable baseline model

10.2 Decision Tree

Decision Trees were used to capture nonlinear relationships in operational logistics data.

Advantages:

- Easy interpretation
- Nonlinear learning capability
- Feature hierarchy visualization

10.3 Random Forest

Random Forest improved predictive performance using ensemble learning.

Advantages:

- Reduced overfitting
- Strong feature importance analysis
- Robust classification performance

10.4 Gradient Boosting

Gradient Boosting improved prediction capability by sequentially correcting weak learners.

Advantages:

- Strong predictive performance
- Handles complex relationships
- Good operational classification

10.5 XGBoost

XGBoost achieved the best overall predictive performance.

Advantages:

- High scalability
 - Excellent predictive accuracy
 - Effective handling of tabular business data
 - Strong regularization support
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11. Model Evaluation

The models were evaluated using multiple classification metrics.

11.1 Accuracy

Measures the overall correctness of predictions.

11.2 Precision

Measures the proportion of predicted delays that were actually delayed.

Precision is important to reduce false operational alerts.

11.3 Recall

Measures the proportion of actual delayed deliveries correctly identified.

Recall is critical because missing true delays can result in major operational losses.

11.4 F1-Score

F1-Score balances Precision and Recall.

11.5 ROC-AUC

ROC-AUC measures the model's ability to distinguish delayed and non-delayed deliveries.

12. Experimental Results

The models achieved the following approximate performance:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.82	0.80	0.84	0.82	0.88
Decision Tree	0.85	0.84	0.86	0.85	0.90
Random Forest	0.91	0.89	0.92	0.90	0.95
Gradient Boosting	0.92	0.90	0.93	0.91	0.96
XGBoost	0.94	0.92	0.95	0.93	0.98

XGBoost achieved the best overall performance and was selected as the final deployment model.

13. Explainable AI Analysis

Explainability is critical for operational trust in logistics systems.

SHAP analysis and feature importance methods were used to identify key contributors to delivery risk.

13.1 Major Risk Drivers

The most influential features included:

- Shipping Pressure Index
- Scheduled Shipment Days
- Order Quantity
- Shipping Mode
- Regional Congestion Score
- Order Complexity Score
- Customer Segment Risk

13.2 Example Risk Explanation

Example High-Risk Shipment:

- High order quantity
- Standard shipping mode
- Congested delivery region
- Tight scheduled delivery window

Predicted Risk Probability: 0.89

14. Risk Classification Framework

Each shipment receives:

- Late Delivery Probability Score
- Risk Category Label

14.1 Risk Categories

Probability Range	Risk Category
0.00 – 0.30	Low Risk
0.31 – 0.70	Medium Risk
0.71 – 1.00	High Risk

This framework supports operational prioritization and intervention planning.

15. Streamlit Dashboard Development

An interactive Streamlit dashboard was developed for real-time operational analytics.

15.1 Dashboard Modules

Delay Risk Overview

Features:

- Overall risk distribution
- High-risk shipment counts
- KPI monitoring
- Trend analysis

Order-Level Prediction

Features:

- Shipment risk probability
- Risk category prediction
- Explainable AI insights

Region and Shipping Analysis

Features:

- Regional risk heatmaps
- Shipping mode comparisons
- Market-level analytics

Operations Action Panel

Features:

- High-risk shipment queue
 - Priority-based intervention list
 - Shipment monitoring system
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16. Business Recommendations

Based on the predictive analytics framework, the following operational recommendations are proposed.

16.1 Risk-Based Shipment Prioritization

High-risk shipments should receive operational priority for faster processing and monitoring.

16.2 Dynamic Shipping Upgrades

Risky standard shipments can be upgraded to express modes proactively.

16.3 Regional Capacity Planning

Additional operational resources should be allocated to high-congestion regions.

16.4 Predictive Alert Systems

Automated alerts should notify operations managers when shipment risk exceeds predefined thresholds.

16.5 Customer Communication Optimization

High-risk customers should receive proactive delivery notifications and status updates.

17. Business Impact

The proposed system provides several operational and business benefits.

17.1 Reduced Shipment Delays

Predictive risk identification allows early operational intervention.

17.2 Lower Operational Costs

Proactive planning reduces emergency logistics expenses.

17.3 Improved Customer Satisfaction

Fewer delays improve customer trust and retention.

17.4 Data-Driven Decision Making

Operations teams gain intelligent insights for shipment planning.

17.5 Enhanced Supply Chain Visibility

Real-time risk monitoring improves operational transparency.

18. Future Scope

Several advanced enhancements can further improve the proposed framework.

18.1 Real-Time Prediction APIs

Deploy prediction services using FastAPI and cloud infrastructure.

18.2 Deep Learning Models

Explore LSTM and Transformer architectures for advanced logistics forecasting.

18.3 IoT Integration

Use sensor-based logistics tracking for real-time shipment intelligence.

18.4 Reinforcement Learning

Optimize route planning and dynamic shipping allocation.

18.5 Route Optimization Systems

Integrate predictive risk models with intelligent route optimization algorithms.

19. Conclusion

This research presented a machine learning–based framework for predicting late delivery risks in global supply chain operations. The proposed system uses operational, financial, customer, geographical, and shipment-related features to estimate delivery delay probability before shipment execution.

Feature engineering techniques significantly improved model performance by capturing operational risk patterns. Multiple machine learning algorithms were evaluated, with XGBoost achieving the best predictive accuracy and ROC-AUC performance.

The integration of explainable AI and Streamlit dashboard analytics transformed the predictive framework into a practical operational intelligence system. The solution enables proactive logistics management, intelligent shipment prioritization, operational transparency, and improved customer service.

The proposed framework demonstrates how machine learning can transform traditional reactive logistics systems into intelligent predictive supply chain ecosystems.

20. References

1. Scikit-learn Documentation
2. XGBoost Official Documentation
3. Streamlit Documentation
4. Research Papers on Supply Chain Analytics
5. Logistics and Transportation Risk Management Studies
6. Machine Learning Applications in Supply Chain Management
7. Predictive Analytics in Logistics Operations
8. Explainable AI Research Publications
9. Global Supply Chain Risk Management Literature
10. Ensemble Learning and Gradient Boosting Research Papers

Appendix

Technologies Used

Component	Technology
Programming Language	Python
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost
Explainable AI	SHAP
Dashboard	Streamlit
Deployment	Streamlit Cloud

GitHub Repository Structure

Late-Delivery-Risk-Prediction/ | |—— data/ |—— models/ |—— README.md
|——research_paper.pdf